

Ali HAJIABADI

Postdoctoral Researcher and Lecturer
ETH Zürich

CONTACT INFORMATION

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RESEARCH INTERESTS

Systems Security, Formal Methods, Computer Architecture, Hardware/Software Co-design, Compilers, Secure Architectures and Software, Microarchitectural Side Channels, Machine Learning Systems Security

EDUCATION

- 2019 - 2024 Doctor of Philosophy in Computer Science
National University of Singapore (NUS), Singapore
Thesis: *"Building Efficient and Secure Processors thorough Hardware/Software Co-design"*
Advisor: Prof. Trevor E. CARLSON
- 2014 - 2019 Bachelor of Science in Computer Engineering
Sharif University of Technology, Tehran, Iran
Thesis: *"High Concurrency Latency Tolerant Register Files for GPUs"*
Advisor: Prof. Hamid SARBAZI-AZAD
- 2009 - 2013 Diploma in Physics and Mathematics Discipline
Shahid Beheshti High School, Birjand, Iran
Affiliated with the National Organization for the Development of Exceptional Talents (NODET)

HONORS & AWARDS

- JUL. 2026 NOTEWORTHY REVIEWER, as a PC member of EuroS&P '26.
- AUG. 2024 DEAN'S GRADUATE RESEARCH EXCELLENCE AWARD.
- JUL. 2024 SCHOOL OF COMPUTING CS PHD THESIS AWARD (Honorable Mention).
- JUN. 2024 BEST PAPER AWARD nomination at DAC '24.
- OCT. 2023 NUSGS RESEARCH INCENTIVE AWARD from School of Computing, NUS (S\$ 2,500 award).
- JAN. 2022 STUDENT TRAVEL AWARD from ASPLOS'22 conference.
- AUG. 2021 RESEARCH ACHIEVEMENT AWARD from School of Computing, NUS.
- MAR. 2020 Invited talk and travel grant for the 2nd Young Architect Workshop at ASPLOS'20, Switzerland.
- FEB. 2019 PRESIDENT'S GRADUATE FELLOWSHIP, the most prestigious doctoral fellowship in Singapore.
- SEP. 2014 Ranked 164th in Iranian National University Entrance Exam among more than 250,000 students.
- 2006/2009 Recognized as talented student in entry exam of NODET for middle school and high school.

PROFESSIONAL EXPERIENCE

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| JUL. 2024 - PRESENT | Postdoctoral Researcher at ETH ZÜRICH, Switzerland
<i>Computer Security (COMSEC) Group</i>
Group Lead: Prof. Kaveh RAZAVI
Research: CPU and DRAM security, novel attacks and defenses for computing systems |
| AUG. 2019 - JUN. 2024 | Graduate Research Assistant at NATIONAL UNIVERSITY OF SINGAPORE, Singapore
<i>NUS Computer Architecture (CompArch) Group</i>
Group Lead: Prof. Trevor E. CARLSON
Research: hardware/software co-design for efficient and secure modern processors |
| JUL. 2016 - JUN. 2019 | Research Assistant at SHARIF UNIVERSITY OF TECHNOLOGY, Tehran, Iran
<i>High Performance Computer Architectures and Networks (HPCAN) Lab</i>
Group Lead: Prof. Hamid SARBAZI-AZAD
Research: GPU register prefetching via HW/SW co-design and compiler optimizations |
| SUMMER 2018 | Research Intern at NATIONAL UNIVERSITY OF SINGAPORE, Singapore
Group Lead: Prof. Trevor E. CARLSON
Research: exploring new implementations for out-of-order commit processors |

PEER-REVIEWED PUBLICATIONS

18. **Chenfei Liu, Ali Hajiabadi, Kaveh Razavi**
ASPLOS'27 *Compression: Impossible – Securing Hardware-Level Memory Compression.*
Proceedings of 32nd ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2027), April 2027.
17. **Ali Hajiabadi, Silvan Niederer, Kaveh Razavi**
CCS'26 *VeriPRAC: Verifiable Formal Contracts for Secure Per Row Activation Counting.*
Proceedings of 33rd ACM Conference on Computer and Communications Security (CCS 2026), November 2026.
16. **Silvan Niederer, Ali Hajiabadi, Kaveh Razavi**
DIMVA'26 *Glasswall: Enforcing User-Defined Data Usage Policies in Functions as a Service.*
Proceedings of 23rd Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA 2026), July 2026. Acceptance rate: 33/151 = 21.8%.
15. **Jean-Claude Graf*, Sandro Rüegg*, Ali Hajiabadi, Kaveh Razavi**
S&P'26 *VMscape: Exposing and Exploiting Incomplete Branch Predictor Isolation in Cloud Environments.*
Proceedings of 47th IEEE Symposium on Security and Privacy (S&P, Oakland 2026), May 2026. Acceptance rate: 118/925 = 12.8%.
14. **Hritvik Taneja, Ali Hajiabadi, Michele Marazzi, Kaveh Razavi, Moinuddin K. Qureshi**
HPCA'26 *MIRZA: Efficiently Mitigating Rowhammer with Randomization and ALERT.*
Proceedings of 32nd IEEE International Symposium on High-Performance Computer Architecture (HPCA 2026), February 2026. Acceptance rate: 119/602 = 19.8%.
13. **Ali Hajiabadi, Michele Marazzi, Kaveh Razavi**
CCS'25 *CHaRM: Checkpointed and Hashed Counters for Flexible and Efficient Rowhammer Mitigation.*
Proceedings of 32nd ACM Conference on Computer and Communications Security (CCS 2025), October 2025. Acceptance rate: 316/2186 = 14.5%.
12. **Silvan Niederer, Sandro Rüegg, Ali Hajiabadi, Kaveh Razavi**
MICRO'25 *One Flew over the Stack Engine's Nest: Practical Microarchitectural Attacks on the Stack Engine.*
Proceedings of 58th IEEE/ACM International Conference on Microarchitecture (MICRO 2025), October 2025. Acceptance rate: 124/597 = 20.8%.
11. **Ali Hajiabadi, Trevor E. Carlson**
ISCA'25 *CASSANDRA: Efficient Enforcement of Sequential Execution for Cryptographic Programs.*
Proceedings of 52nd ACM/IEEE International Symposium on Computer Architecture (ISCA 2025), June 2025. Acceptance rate: 132/570 = 23.1%.
10. **Yun Chen*, Ali Hajiabadi*, Romain Poussier, Yaswanth Tavva, Andreas Diavastos, Shivam Bhasin, Trevor E. Carlson**
TACO'25 *PARADISE: Criticality-Aware Instruction Reordering for Power Attack Resistance.*
In ACM Transactions on Architecture and Code Generation (TACO), 2024.
*Joint first-authors with equal contribution.
9. **Ali Hajiabadi, Trevor E. Carlson**
DAC'24 *CONJURING: Leaking Control Flow via Speculative Fetch Attacks.*
Proceedings of 61st ACM/IEEE Design Automation Conference (DAC 2024), June 2024. Acceptance rate: 337/1465 = 23.0%.
★ *Best Paper Award Nominee* (5/337 = 1.5%).
8. **Ali Hajiabadi, Archit Agarwal, Andreas Diavastos, Trevor E. Carlson**
DAC'24 *LEVISO: Efficient Compiler-Informed Secure Speculation.*
Proceedings of 61st ACM/IEEE Design Automation Conference (DAC 2024), June 2024. Acceptance rate: 337/1465 = 23.0%.
7. **Yun Chen*, Ali Hajiabadi*, Trevor E. Carlson**
HPCA'24 *GADGETSPINNER: A New Transient Execution Primitive using the Loop Stream Detector.*
Proceedings of 30th IEEE International Symposium on High-Performance Computer Architecture (HPCA 2024), March 2024. Acceptance rate: 75/410 = 18.3%.
*Joint first-authors with equal contribution.
6. **Yun Chen, Ali Hajiabadi, Lingfeng Pei, Trevor E. Carlson**
HPCA'24 *PREFETCHX: Cross-Core Cache-Agnostic Prefetcher-Based Side-Channel Attacks.*
Proceedings of 30th IEEE International Symposium on High-Performance Computer Architecture (HPCA 2024), March 2024. Acceptance rate: 75/410 = 18.3%.

5. | Arash Pashrashid, **Ali Hajiabadi**, Trevor E. Carlson
 ICCAD'23 | *HIDFIX: Efficient Mitigation of Cache-based Spectre Attacks through Hidden Rollbacks*.
 Proceedings of 42nd IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2023),
 November 2023. Acceptance rate: 172/768 = 22.4%.
4. | Arash Pashrashid, **Ali Hajiabadi**, Trevor E. Carlson
 ICCAD'22 | *Fast, Robust and Accurate Detection of Cache-based Spectre Attack Phases*.
 Proceedings of 41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2022),
 November 2022. Acceptance rate: 132/586 = 22.5%.
3. | **Ali Hajiabadi**, Andreas Diavastos, Trevor E. Carlson
 ASPLOS'21 | *NOREBA: A Compiler-Informed Non-speculative Out-of-Order Commit Processor*.
 Proceedings of 26th ACM International Conference on Architectural Support for Programming
 Languages and Operating Systems (ASPLOS 2021), April 2021. Acceptance rate: 75/398 = 18.8%.
2. | Mohammad Sadrosadati, Amirhossein Mirhosseini, **Ali Hajiabadi**, Seyed Borna Ehsani, Hajar Fala-
 TOCS'21 | *Highly Concurrent Latency-tolerant Register Files for GPUs*.
 In ACM Transactions on Computer Systems (TOCS), 2021.
1. | Harish Patil, Alexander Isaev, Wim Heirman, Alen Sabu, **Ali Hajiabadi**, Trevor E. Carlson
 CGO'21 | *ELFies: Executable Region Checkpoints for Performance Analysis and Simulation*.
 Proceedings of 19th IEEE International Symposium on Code Generation and Optimization (CGO
 2021), March 2021. Acceptance rate: 31/89 = 34.8%.

TEACHING EXPERIENCE

► ETH Zürich, Switzerland

- FALL 2026 **Lecturer**, Computer Security (BSc) [**New: RISC-V edition**]
 SPRING 2026 **Co-lecturer**, Computer Engineering (BSc), *with Prof. Kaveh Razavi and Dr. Patrick Jattke*
 FALL 2025 **Lecturer**, Computer Security (BSc) [**New Course**]
 SPRING 2025 **Lecturer**, Capture the Flag – Introduction to Cybersecurity (P&S BSc) [**New Course**]
 SPRING 2025 **Co-lecturer**, Computer Engineering (BSc), *with Prof. Kaveh Razavi*

► National University of Singapore, Singapore

- SPRING 2020 **Teaching Assistant** (*tutorial instructor*), CS2106 Introduction to Operating Systems (BSc)
 & SPRING 2021 *Lecturer: Prof. Djordje Jevdjic*

► Sharif University of Technology, Tehran, Iran

- SPRING 2017 **Teaching Assistant** (*assignments & projects*), CE323 Computer Architecture (BSc)
Lecturer: Prof. Hamid Sarbazi-Azad
 FALL 2017 **Teaching Assistant** (*tutorial instructor, assignments & projects*), CE453 Real-Time Systems (BSc)
 & FALL 2018 *Lecturer: Prof. Amirhossein Jahangir*

PROFESSIONAL SERVICES

- Program Committee** S&P ('27), uASC ('27, '26), HPCA ('27, '26), MICRO ('26, '25), ISCA ('26), EuroS&P ('27, '26),
 AsiaCCS ('26)
Sub-reviewer USENIX Security ('25)
Reviewer (journals) IEEE CAL, ACM TACO, ACM TRETs
Session Chair uASC'26 (*Brave New Defense*), MICRO'25 (*Security and Privacy - Memory*)
Shadow PC EuroSys ('24, '23)
Student Volunteer PLDI ('21)

TALKS

- NOV. 2025 **The Hidden and Ugly Faces of Spectre: Unexplored and Unmitigated Threat Models**
IC, EPFL, Lausanne, Switzerland.
 OCT. 2025 **CHaRM: Checkpointed and Hashed Counters for Flexible and Efficient Rowhammer Mitigation**
Conference on Computer and Communication Security (CCS 2025), Taipei, Taiwan.
 JUN. 2025 **CASSANDRA: Efficient Enforcement of Sequential Execution for Cryptographic Programs**
Int. Symposium on Computer Architecture (ISCA 2025), Tokyo, Japan.
 MAR. 2024 **Will CPUs Be Free of Spectre? Dark Side and Light Side of the Battle**
ETH Zürich, COMSEC Group, Zürich, Switzerland.

MAR. 2024	GADGETSPINNER: A New Transient Execution Primitive using the Loop Stream Detector <i>Int. Symposium on High-Performance Computer Architecture (HPCA 2024)</i> , Edinburgh, Scotland.
AUG. 2021	NOREBA: A Compiler-Informed Non-speculative Out-of-Order Commit Processor <i>Computing Research Week, School of Computing (NUS)</i> , Virtual.
APR. 2021	NOREBA: A Compiler-Informed Non-speculative Out-of-Order Commit Processor <i>Int. Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2021)</i> , Virtual.
FEB. 2021	Accelerating HPC applications with Out-of-Order Commit Processors <i>Free and Open source Software Developers' European Meeting (FOSDEM 2021)</i> , HPC, Big Data, and Data Science track, Virtual.
MAR. 2020	Speculation-Free Out-of-Order Commit <i>2nd Young Architect Workshop at the 25th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020)</i> , Virtual.

RESEARCH SUPERVISION

MAR. 2026 - JUN. 2026	<i>Yves Rüetschi</i>	BSc thesis, ETH Zürich	
FEB. 2026 - JUN. 2026	<i>Tomas Jossen</i>	BSc thesis, ETH Zürich	
FEB. 2026 - PRESENT	<i>Jonas Buchholz</i>	MSc thesis, ETH Zürich	<i>Embargo</i>
JAN. 2026 - PRESENT	<i>Quentin Bordier</i>	PhD thesis, COMSEC, ETH Zürich	<i>Embargo</i>
MAR. 2025 - MAY 2026	<i>Jean-Claude Graf</i>	MSc thesis, COMSEC, ETH Zürich	<i>1 publication (SGP'26)</i>
MAR. 2025 - MAY 2026	<i>Sandro Rüegge</i>	PhD thesis, COMSEC, ETH Zürich	<i>1 publication (SGP'26)</i>
MAR. 2025 - SEP. 2025	<i>Maximilian Mosler</i>	MSc thesis, ETH Zürich	<i>Embargo</i>
MAR. 2025 - JUN. 2025	<i>Jonas Buchholz</i>	Semester project, ETH Zürich	<i>Embargo</i>
DEC. 2024 - JUN. 2025	<i>Chenfei Liu</i>	MSc thesis, ETH Zürich	<i>1 publication (ASPLOS'27)</i>
SEP. 2024 - FEB. 2025	<i>Mounir Raki</i>	Semester project, ETH Zürich	
JUL. 2024 - PRESENT	<i>Silvan Niderer</i>	PhD thesis, COMSEC, ETH Zürich	<i>2 publications (MICRO'25, DIMVA'26)</i>
JAN. 2021 - DEC. 2023	<i>Arash Pashrashid</i>	PhD thesis, NUS CompArch	<i>2 publications (ICCAD'23, ICCAD'22)</i>
JUL. 2021 - JUL. 2023	<i>Archit Agarwal</i>	RA, NUS CompArch	<i>1 publication (DAC'24)</i>
SEP. 2020 - SEP. 2021	<i>Vernon Pang</i>	BSc final-year project, NUS	
AUG. 2020 - MAR. 2024	<i>Yun Chen</i>	PhD thesis, NUS CompArch	<i>3 publications (2×HPCA'24, TACO)</i>